Geometry-Dependent Classification of Reduced Potentials in Einstein–Cartan Gravity

Muacca

Version: v3
First release: 2026-01-11
Revised: 2026-10-25

Abstract

We compare the Einstein–Cartan (EC) action with a constant complete Nieh–Yan (NY) coupling restricted to static homogeneous geometries on $S^3 \times S^1$, $T^3 \times S^1$, and compact $\mathrm{Nil}^3 \times S^1$. A metric-compatible connection carries axial and vector torsion. Specifying the connection and local invariants separately from lattices, periods, and integration volumes fixes the global normalization of each local calculation.

On the ansatz, the two contributions to the complete NY density coincide: $T^a \wedge T_a - e^a \wedge e^b \wedge R_{ab} = 0$. The reduced potential of the complete action is therefore independent of the NY coupling θ. In the one-variable classification with torsion coefficients and circle circumference fixed, S^3 has a strict radius minimum without a barrier for $V \neq 0$, $|\eta| < 2$. The complete action has no strict positive-radius minimum on isotropic or fixed-shape torus expansion, or on uniform Nil expansion. Varying one torus radius with the other two fixed is a distinct problem, whose minimum condition is also stated.

Diagnostic actions adding only the torsion-side or positive curvature-side NY density coincide on the ansatz. They can have barrier-separated minima from the mixed term, but these are not results for the complete NY action. The classification follows analytically from cubic polynomials, with boundaries, degeneracies, and numerical limitations distinguished.

These are off-shell classifications along fixed variation paths. A radius minimum does not ensure stationarity under full variations including the independent connection, torsion, and circle circumference, Lorentzian dynamical stability, or physical masses and decay rates. For standard source-free EC gravity without a cosmological constant and with a constant NY coupling, a separate nonexistence obstruction applies to smooth nondegenerate full vacuum solutions on closed real Euclidean $S^3 \times S^1$.

Contents

1 Introduction 3

2 Action, connection, and variations 4

3 Global geometry and reduced potentials 7

4 Analytic classification on the positive-radius domain 11

5 Extremum values and their interpretation 16

6 Representative parameters and scaling 17

7 Variation scope and physical interpretation 19

8 Conclusions and open questions 20

A Connection components and the code-input dictionary 22

B Checks of the connection, invariants, and classification 23

C Numerical classification and input domains 24

D Displaying potentials and phase diagrams 26

E Anisotropic torus and variation paths 27

F Code, data, and reproducibility 28

G Curvature in a noncoordinate frame 29

1 Introduction

1.1 Question and scope

We study reduced potentials in Einstein–Cartan (EC) gravity with a constant Nieh–Yan (NY) coupling on static, homogeneous Euclidean geometries $S^3 \times S^1$, $T^3 \times S^1$, and compact $\mathrm{Nil}^3 \times S^1$. The connection is metric compatible and admits axial and vector torsion. The question is how the extrema of a specified one-parameter restriction of the action depend on the geometry, the torsion parameters, and the chosen action variant.

Throughout this paper, a *phase* means an extremum class on the open positive-radius domain. It denotes neither a thermodynamic phase, a transition between topologies, nor a classification of solutions of the full field equations. We define $U = -S$ after integration over all four dimensions. Without a kinetic term, a time prescription, constraints, and a physical unit dictionary, this quantity cannot be identified with an energy or a particle mass.

1.2 Torsion and comparison of geometries

The same torsion ansatz is used on the three spatial geometries. Local structure coefficients and curvature are specified separately from the global quotient, coordinate periods, and integration volume. The EC framework and the NY identity provide the underlying geometric setting [1, 3, 2].

In the preceding version of this paper and its computational implementation, the input array was called T. With the Cartan equation and index order used here, the torsion reconstructed from that connection is the negative of this array. We denote the input array by τ and use T for geometric torsion, so that $\tau = -T$. The parameters V, η in the main text describe T. Appendix A gives the connection dictionary, and Appendix C gives the input-domain dictionary.

This dictionary preserves the formulas used to construct the contortion and connection from the code input: simultaneous conversion of the array and its parameters describes the same connection. It is distinct from the correction of the NY contraction coefficient, which changes the reduced potentials.

1.3 Principal conclusions

On the ansatz considered here, the TT and positive-REE densities coincide pointwise. Their difference, the complete NY density, vanishes. Consequently, the complete action, denoted FULL, has no θ-dependent phase boundary. For fixed torsion parameters and circle length, S^3 can support a strict radius minimum through its negative linear term and the positive cubic vector-torsion term. The isotropic or fixed-shape T^3 path and the uniform Nil path do not support such a minimum in FULL.

The separate TT and positive-REE actions are diagnostic comparisons. Their mixed term is proportional to $V\eta\theta$, and they can have barriers. Those barriers are not properties of FULL. The comparison is geometry dependent: the global quotient and the path of variation are part of the problem. It does not establish vacuum selection by topology.

1.4 Main points of this revision

1. The coefficient in the REE contraction is $1/2$, replacing $1/4$; the TT coefficient remains $1/4$. For this ansatz the complete NY density therefore vanishes.

2. Geometric torsion T is the primary variable. The relation $\tau = -T$ preserves the connection associated with the computational input and is independent of the NY coefficient correction.

3. Global normalization is fixed explicitly: the round-sphere radius is $a = r/2$, L is the circle circumference, $\mathrm{Vol}(S^3 \times S^1) = \pi^2 L r^3/4$, and the spatial torus angles have period 2π, giving volume $(2\pi)^3 L R_1 R_2 R_3$. For Nil, the lattice $(a, b, c) = (2\pi, 2\pi, 4\pi^2)$ realizes the volume $(2\pi)^4 L R^3$.

4. Claims of θ-dependent stabilization and phase boundaries in FULL are withdrawn. Diagnostic actions are distinguished from FULL, and radius minima are distinguished from full vacuum solutions. Potential heights are not directly interpreted as masses or decay rates.

1.5 Organization

Section 2 defines the action, conventions, and variations. Section 3 fixes the global geometries and reduced potentials. Sections 4–6 give the analytic classification, extremum measures, and representative parameter choices. Section 7 states the limits of physical interpretation, and Section 8 summarizes the conclusions. Appendices A–G provide the torsion dictionary, verification criteria, numerical input and output conventions, visualization conventions, torus variation paths, code access, and the noncoordinate curvature formula.

2 Action, connection, and variations

2.1 Coframe and index conventions

We use the four-dimensional Euclidean metric δ_{ab}, with $a, b, c = 0, 1, 2, 3$ and spatial indices $i, j, k = 0, 1, 2$. For an orthonormal coframe e^a and its dual frame E_a, $\varepsilon_{0123} = +1$ and $\mathrm{vol}_4 = e^0 \wedge e^1 \wedge e^2 \wedge e^3$. The structure coefficients have the positive-sign convention

$$de^a = \frac{1}{2} C^a{}_{bc} e^b \wedge e^c, \qquad [E_b, E_c] = -C^a{}_{bc} E_a, \qquad \omega^a{}_b = \Gamma^a{}_{bc} e^c. \tag{M2-01}$$

The last index of the connection coefficient is the differentiation direction. Metric compatibility means $\Gamma_{abc} = -\Gamma_{bac}$. The components of the torsion and curvature two-forms are defined by

$$\begin{aligned} T^a &= de^a + \omega^a{}_b \wedge e^b = \frac{1}{2} T^a{}_{bc} e^b \wedge e^c, \\ R^a{}_b &= d\omega^a{}_b + \omega^a{}_c \wedge \omega^c{}_b = \frac{1}{2} R^a{}_{bcd} e^c \wedge e^d. \end{aligned} \tag{M2-02}$$

2.2 Reconstructing the connection

The connection constructed from geometric torsion is

$$\begin{aligned} \Gamma_{abc} &= \Gamma^{\mathrm{LC}}_{abc} + K_{abc}[T], \\ \Gamma^{\mathrm{LC}}_{abc} &= \frac{1}{2}(C_{abc} + C_{cba} - C_{bac}), \\ K_{abc}[T] &= -\frac{1}{2}(T_{abc} + T_{bca} - T_{cab}). \end{aligned} \tag{M2-03}$$

The sign follows from (M2-01), (M2-02), and the index order. Substitution in $T^a{}_{bc} = C^a{}_{bc} + \Gamma^a{}_{cb} - \Gamma^a{}_{bc}$ recovers the prescribed T. The curvature includes the noncoordinate term:

$$R^a{}_{bcd} = E_c(\Gamma^a{}_{bd}) - E_d(\Gamma^a{}_{bc}) + \Gamma^a{}_{ec}\Gamma^e{}_{bd} - \Gamma^a{}_{ed}\Gamma^e{}_{bc} + \Gamma^a{}_{be}C^e{}_{cd}. \tag{M2-04}$$

Frame components are constant at each fixed scale in this ansatz, so the first two derivative terms vanish. Comparing different geometries by varying the scale is a different operation from differentiation in a frame direction.

2.3 Torsion ansatz

We specify geometric torsion as the sum of axial and vector parts. Here $s = r$ for S^3, $s = R_1$ for T^3, and $s = R$ for Nil. The axial part has spatial indices only:

$$\begin{aligned} T^{\mathrm{ax}}_{ijk} &= \frac{2\eta}{s}\varepsilon_{ijk}, & T^{\mathrm{ax}}_{ab3} = T^{\mathrm{ax}}_{a3b} = T^{\mathrm{ax}}_{3ab} &= 0, \\ T^{\mathrm{vec}}_{abc} &= \frac{1}{3}(\delta_{ac}V_b - \delta_{ab}V_c), & V_a &= (0,0,0,V), \\ T_{abc} &= T^{\mathrm{ax}}_{abc} + T^{\mathrm{vec}}_{abc}. \end{aligned} \tag{M2-05}$$

The parameters η, V are signed real numbers, and $T^\lambda{}_{\mu\lambda} = V_\mu$. AX denotes $V = 0$, VT denotes $\eta = 0$, and MX allows both parts. Zero values are included in the formulas. Restricting V to nonnegative values is a choice of input domain, not a geometric definition.

We hold η, V fixed in the reduced variation. Thus the axial frame component $2\eta/s$ changes with scale. This differs both from holding all torsion frame components fixed and from freely varying an independent connection.

2.4 Complete Nieh–Yan term and diagnostic actions

The NY four-form and its component densities are

$$\begin{aligned} \mathcal{N} &= d(e^a \wedge T_a) = T^a \wedge T_a - e^a \wedge e^b \wedge R_{ab} = n_{\mathrm{FULL}}\,\mathrm{vol}_4, \\ n_{\mathrm{TT}} &= \frac{1}{4}\varepsilon^{abcd}T^e{}_{ab}T_{ecd}, \\ n_{\mathrm{REE}} &= \frac{1}{2}\varepsilon^{abcd}R_{abcd}, \\ n_{\mathrm{FULL}} &= n_{\mathrm{TT}} - n_{\mathrm{REE}}. \end{aligned} \tag{M2-06}$$

The TT factor $1/4$ comes from the two torsion two-forms, while the REE factor $1/2$ comes from the single curvature two-form. Pointwise on the present ansatz,

$$n_{\mathrm{TT}} = n_{\mathrm{REE}} = -\frac{4V\eta}{s}, \qquad n_{\mathrm{FULL}} = 0. \tag{M2-07}$$

For $\kappa^2 > 0$ and a spacetime-constant θ, the complete action is

$$S_{\mathrm{FULL}} = \int_M \frac{\mathcal{R}_{\mathrm{EC}}}{2\kappa^2}\,\mathrm{vol}_4 + \theta\int_M \mathcal{N}, \qquad U_{\mathrm{FULL}} = -S_{\mathrm{FULL}}. \tag{M2-08}$$

For comparison we define $S_{\mathrm{TT}} = \int_M(\mathcal{R}_{\mathrm{EC}}/(2\kappa^2) + \theta n_{\mathrm{TT}})\mathrm{vol}_4$ and $S_{\mathrm{REE}} = \int_M(\mathcal{R}_{\mathrm{EC}}/(2\kappa^2) + \theta n_{\mathrm{REE}})\mathrm{vol}_4$, with $U = -S$ in each case. The REE diagnostic adds the curvature-side density

with a *positive* sign. It therefore differs in sign from isolating the negative component of the complete NY term. With this definition, $U_{\mathrm{TT}} = U_{\mathrm{REE}}$ on the ansatz.

If $e^a \wedge T_a$ is globally defined and smooth on a closed manifold, Stokes' theorem makes the complete NY integral zero. Here its density also vanishes pointwise. This conclusion is not transferred unchanged to problems with boundaries, singularities, or a nonconstant coupling.

2.5 Integration, dimensions, and variation domain

The common circle coordinate is $u \in [0, 1)$, with $e^3 = L\,du$. Thus $L > 0$ is its actual circumference, and the action includes integration over this direction. In length units, η is dimensionless, V has dimension inverse length, and κ^2 has dimension length squared. For the action in (M2-08) to be dimensionless, θ has dimension inverse length squared; $\kappa^2\theta$ is dimensionless. Setting $\kappa = L = 1$ in numerical examples selects reference units. It does not identify U with a Planck mass.

The main variation paths change r on S^3, R on Nil, and either $R_i = r$ or $R_2 = \alpha R_1$, $R_3 = \beta R_1$ on T^3. The positive shape ratios α, β and $\eta, V, L, \kappa, \theta$ are fixed. Appendix E separately treats the torus path with R_2, R_3 fixed.

Stationarity of the one-variable action after substitution does not replace the Euler–Lagrange equations for the independent connection, all coframe components, L, or the torsion coefficients. We call the resulting extrema radius minima or reduced minima, distinguishing them from full vacuum solutions.

2.6 Figures and summary tables

The construction, curves, or classification are illustrated in Figure 1.

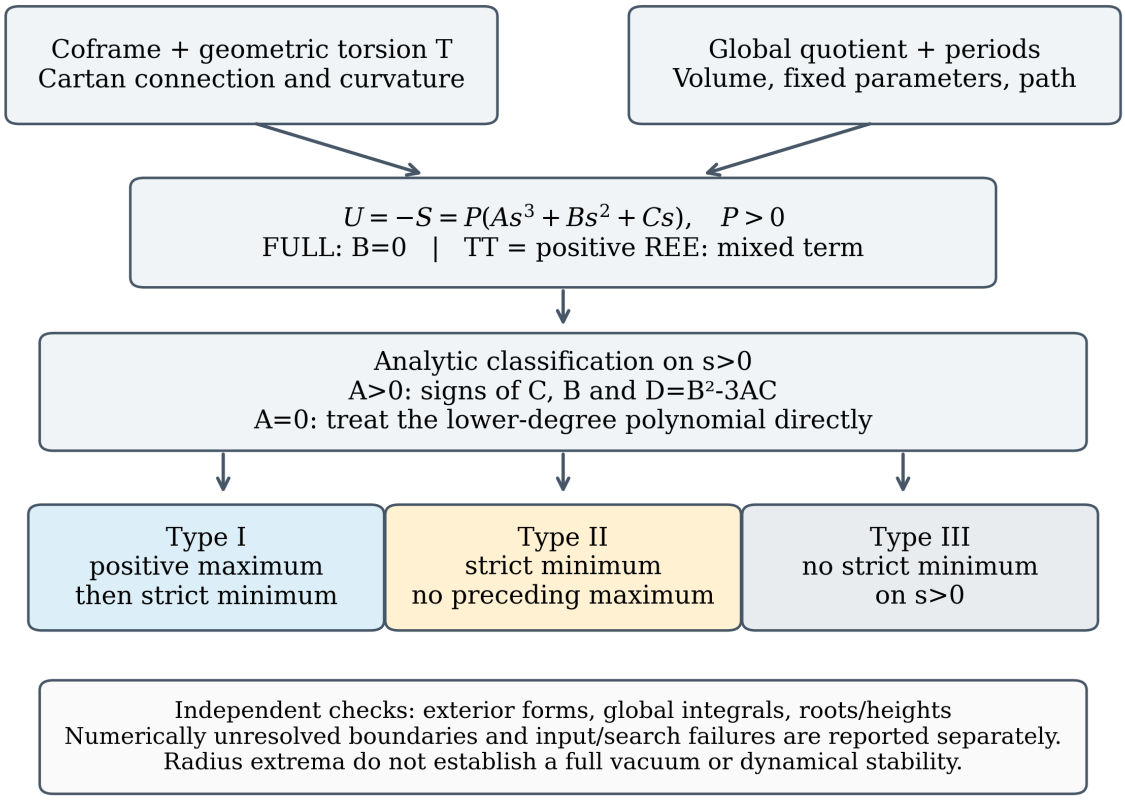


Figure 1: From action to radius classification. Local connections and global integration data are specified separately. Classification is analytic on positive scales; numerical uncertainty and input failures are not physical phases.

3 Global geometry and reduced potentials

3.1 The three-sphere: structure coefficients and physical radius

Identify S^3 with $SU(2)$ and take Euler angles $\vartheta \in [0, \pi]$, $\varphi \in [0, 2\pi)$, $\psi \in [0, 4\pi)$. Use the left-invariant forms

$$\begin{aligned} \lambda^0 &= \cos\psi\, d\vartheta + \sin\psi \sin\vartheta\, d\varphi, \\ \lambda^1 &= -\sin\psi\, d\vartheta + \cos\psi \sin\vartheta\, d\varphi, \\ \lambda^2 &= d\psi + \cos\vartheta\, d\varphi, \qquad e^i = -\frac{r}{4}\lambda^i. \end{aligned} \tag{M3-01}$$

At the boundaries, $(\varphi + 2\pi, \psi) \sim (\varphi, \psi + 2\pi)$, together with the usual $SU(2)$ identifications at the polar coordinate degeneracies. These angles are not three independent circles. The coframe gives $C^i{}_{jk} = 4\varepsilon_{ijk}/r$, and the actual radius of the round sphere is $a = r/2$.

The positive volume measure compatible with the coframe orientation is $r^3 \sin\vartheta\, d\vartheta\, d\varphi\, d\psi/64$. The sign of the angular coordinate ordering is distinguished from the coframe orientation. The

volumes and local invariants are

$$\begin{aligned} \mathrm{Vol}(S^3) &= \frac{\pi^2 r^3}{4} = 2\pi^2 a^3, \\ \mathrm{Vol}(S^3 \times S^1) &= \frac{\pi^2 L r^3}{4}, \\ \mathcal{R}_{\mathrm{LC}} &= \frac{24}{r^2} = \frac{6}{a^2}, \\ \mathcal{R}_{\mathrm{EC}} &= \frac{24 - 6\eta^2}{r^2} - \frac{2}{3}V^2. \end{aligned} \tag{M3-02}$$

It follows that

$$\begin{aligned} U^{S^3}_{\mathrm{FULL}}(r) &= \frac{\pi^2 L}{12\kappa^2}\left[V^2 r^3 + (9\eta^2 - 36)r\right], \\ U^{S^3}_{\mathrm{TT}}(r) = U^{S^3}_{\mathrm{REE}}(r) &= \frac{\pi^2 L}{12\kappa^2}\left[V^2 r^3 + 12V\eta\kappa^2\theta r^2 + (9\eta^2 - 36)r\right]. \end{aligned} \tag{M3-03}$$

To express the axial component as $2\chi/a$ in terms of the actual radius, we also transform $\chi = \eta/2$. The same fixed-χ path gives

$$U^{S^3}_{\mathrm{FULL}}(a) = \frac{2\pi^2 L}{3\kappa^2}\left[V^2 a^3 + 9(\chi^2 - 1)a\right]. \tag{M3-04}$$

For each diagnostic action, add $12V\chi\kappa^2\theta a^2$ inside the brackets. Replacing the radius symbol alone while keeping the definition of the axial coefficient unchanged is not this transformation.

3.2 The three-torus: periods and expansion paths

Let $x^i \in [0, 2\pi)$ and $e^{i-1} = R_i\, dx^i$. The positive coframe coefficients R_i correspond to actual circumferences $\ell_i = 2\pi R_i$. Structure coefficients and Levi–Civita curvature vanish, but the EC curvature with torsion need not vanish:

$$\mathrm{Vol}(T^3 \times S^1) = (2\pi)^3 L R_1 R_2 R_3, \qquad \mathcal{R}_{\mathrm{EC}} = -\frac{6\eta^2}{R_1^2} - \frac{2}{3}V^2 \tag{M3-05}$$

Along the isotropic path $R_i = r$,

$$\begin{aligned} U^{T^3}_{\mathrm{FULL}}(r) &= \frac{8\pi^3 L}{3\kappa^2}\left[V^2 r^3 + 9\eta^2 r\right], \\ U^{T^3}_{\mathrm{TT}}(r) = U^{T^3}_{\mathrm{REE}}(r) &= \frac{8\pi^3 L}{3\kappa^2}\left[V^2 r^3 + 12V\eta\kappa^2\theta r^2 + 9\eta^2 r\right]. \end{aligned} \tag{M3-06}$$

For fixed positive shape ratios $R_2 = \alpha R_1$, $R_3 = \beta R_1$, set $r = R_1$ and multiply the potential by $\alpha\beta$. The radius classification is unchanged, but stability under shape variations does not follow. Writing the axial component as $2\zeta/\ell_1$ requires $\zeta = 2\pi\eta$.

3.3 Nil geometry: compact quotient and lattice

Define the Heisenberg group H by $(x, y, z)(x', y', z') = (x + x', y + y', z + z' + xy')$, with coframe

$$e^0 = R\, dx, \qquad e^1 = R\, dy, \qquad e^2 = R(dz - x\, dy), \qquad R > 0. \tag{M3-07}$$

The nonzero structure coefficients are $C^2{}_{01} = -1/R$ and $C^2{}_{10} = 1/R$. We use the left quotient $\Gamma\backslash H$, where $\Gamma = \{(2\pi j, 2\pi k, 4\pi^2 l) : j, k, l \in \mathbb{Z}\}$. For general lattice periods (a, b, c), closure under the group product requires $ab/c \in \mathbb{Z}$. Our choice $(a, b, c) = (2\pi, 2\pi, 4\pi^2)$ has $ab/c = 1$. The coframe descends to the quotient and the spatial manifold is compact.

A fundamental integration region is $x, y \in [0, 2\pi)$, $z \in [0, 4\pi^2)$, with the twisted face identifications

$$(x, y, z) \sim (x + 2\pi, y, z + 2\pi y), \quad (x, y, z) \sim (x, y + 2\pi, z), \quad (x, y, z) \sim (x, y, z + 4\pi^2). \tag{M3-08}$$

This is not a product torus with three independently periodic coordinates. The volume form is $R^3 dx \wedge dy \wedge dz$. The two base circumferences are $2\pi R$, and the central fiber circumference is $4\pi^2 R$. We hold this lattice fixed while varying R:

$$\begin{aligned} \mathrm{Vol}(\mathrm{Nil}^3 \times S^1) &= (2\pi)^4 L R^3, \\ \mathcal{R}_{\mathrm{LC}} &= -\frac{1}{2R^2}, \\ \mathcal{R}_{\mathrm{EC}} &= -\frac{1/2 + 6\eta^2}{R^2} - \frac{2}{3}V^2 \end{aligned} \tag{M3-09}$$

The reduced potentials are

$$\begin{aligned} U^{\mathrm{Nil}}_{\mathrm{FULL}}(R) &= \frac{4\pi^4 L}{3\kappa^2}\left[4V^2R^3 + (36\eta^2 + 3)R\right], \\ U^{\mathrm{Nil}}_{\mathrm{TT}}(R) = U^{\mathrm{Nil}}_{\mathrm{REE}}(R) &= \frac{4\pi^4 L}{3\kappa^2}\left[4V^2R^3 + 48V\eta\kappa^2\theta R^2 + (36\eta^2 + 3)R\right]. \end{aligned} \tag{M3-10}$$

Local structure coefficients alone do not determine the lattice. The quotient above is part of our definition of the geometry, and provides a concrete global realization of the volume $(2\pi)^4 L R^3$.

3.4 Common structure

On all three geometries, $T_{abc}T^{abc} = 24\eta^2/s^2 + 2V^2/3$, and the NY densities obey (M2-07). Denoting the common scale by r, each potential has the form $U = P(Ar^3 + Br^2 + Cr)$ with $P > 0$. FULL has $B = 0$, whereas the diagnostic actions have $B \propto V\eta\theta$. The coefficient C contains the contribution of the Levi–Civita curvature.

A positive constant change of volume normalization multiplies U, its derivatives, and its heights by the same factor at fixed input, without changing positive-radius extremum locations or their classification. Conversion to an actual radius or circumference also requires transforming the horizontal coordinate and the axial parameter.

3.5 Figures and summary tables

See Table 1, Table 2 for the conditions and values.

The construction, curves, or classification are illustrated in Figure 2, Figure 3.

Table 1: Local curvature and global normalization. The S^1 coordinate has unit period and circumference L. Nil face identifications are specified in Section 3.

Space	Scale / periods	$\mathcal{R}_{\mathrm{LC}}$	Four-volume
S^3	$a = r/2$	$24/r^2$	$\pi^2 L r^3/4$
T^3	$\ell_i = 2\pi R_i$	0	$(2\pi)^3 L R_1 R_2 R_3$
Nil^3	$(a,b,c) = (2\pi, 2\pi, 4\pi^2)$	$-1/(2R^2)$	$(2\pi)^4 L R^3$

Table 2: Coefficients of $U = P(As^3 + Bs^2 + Cs)$, with $s = r$ for the sphere and isotropic torus, and $s = R$ for Nil. FULL has $B = 0$; B_{diag} applies to TT and positive REE.

Space	P	A	B_{diag}	C
S^3	$\pi^2 L/(12\kappa^2)$	V^2	$12V\eta\kappa^2\theta$	$9\eta^2 - 36$
T^3	$8\pi^3 L/(3\kappa^2)$	V^2	$12V\eta\kappa^2\theta$	$9\eta^2$
Nil^3	$4\pi^4 L/(3\kappa^2)$	$4V^2$	$48V\eta\kappa^2\theta$	$36\eta^2 + 3$

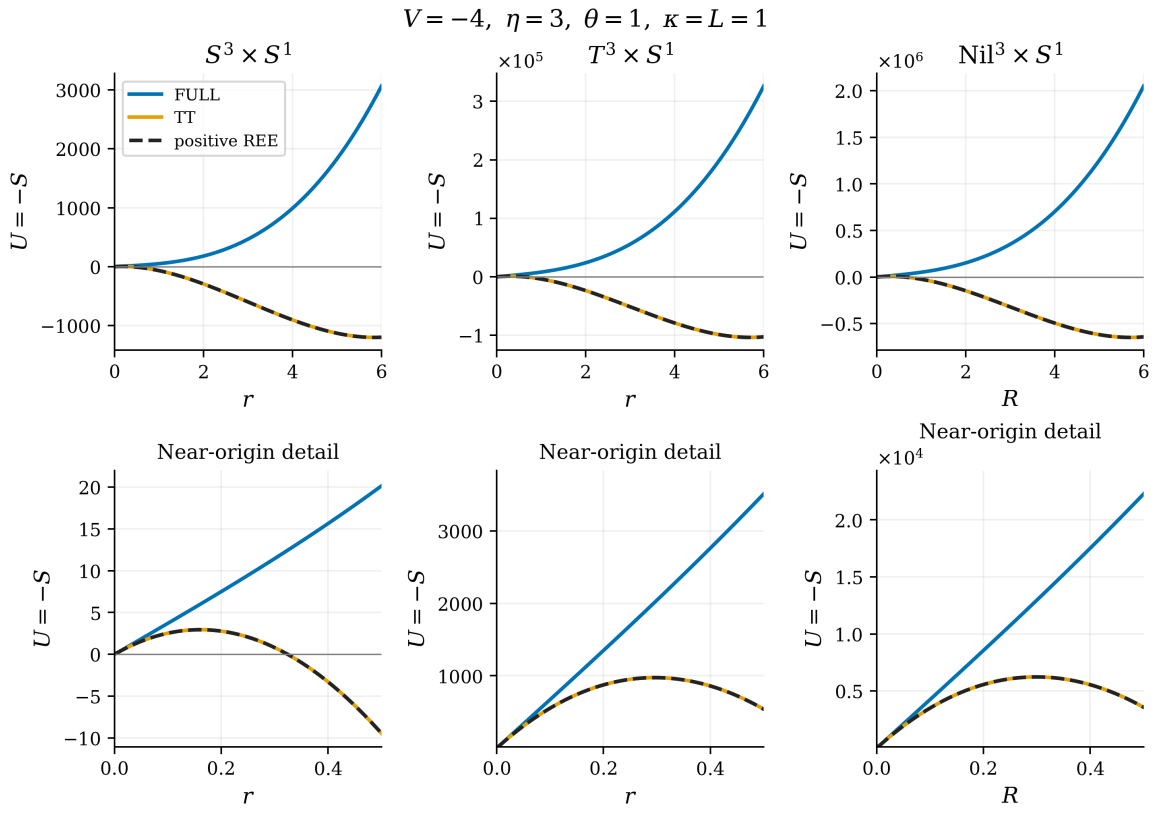


Figure 2: Action variants at $V = -4, \eta = 3, \theta = \kappa = L = 1$. The bottom row magnifies the near-origin region. FULL is Type III in every geometry; TT and positive REE coincide and are Type I. Horizontal axes are coframe scales r (S^3, T^3) or R (Nil); the round-sphere radius is $a = r/2$. Vertical scales use the volumes in Table 1.

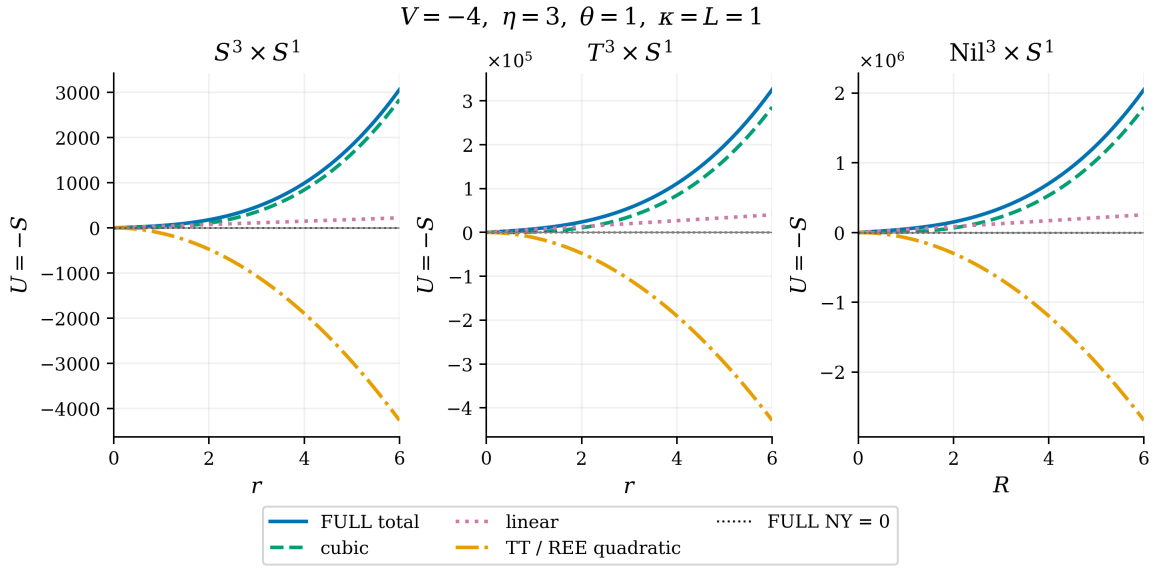


Figure 3: Term decomposition for the inputs of Figure 2. FULL is the sum of cubic and linear terms; its complete NY term is zero. The orange curve is the quadratic contribution to TT or positive REE and is absent from FULL.

4 Analytic classification on the positive-radius domain

4.1 Definitions and cubic criterion

The domain is $r > 0$. Type I has a positive-radius local maximum followed by a strict local minimum. Type II has a strict local minimum without a preceding positive-radius maximum. Type III has no strict local minimum; it includes monotone, runaway, flat, and stationary-inflection cases. A numerical failure is not a physical Type III result.

For $U = P(Ar^3 + Br^2 + Cr)$ with $P, A > 0$, define

$$U' = P(3Ar^2 + 2Br + C), \qquad D = B^2 - 3AC, \qquad r_\pm = \frac{-B \pm \sqrt{D}}{3A}. \tag{M4-01}$$

For $D > 0$, the second derivatives at the two roots are $U''(r_\pm) = \pm 2P\sqrt{D}$, so a positive r_+ is the strict-minimum candidate. The signs of the roots give necessary and sufficient conditions. The complete criterion is:

- $C < 0$: Type II. The product of the roots, $C/(3A)$, is negative, so only r_+ is positive; no additional $B < 0$ condition is needed.
- $C = 0$ and $B < 0$: Type II, with $r_+ = -2B/(3A) > 0$; the root at $r = 0$ is excluded.
- $C > 0$, $B < 0$, and $D > 0$: Type I, with $0 < r_- < r_+$.
- Otherwise: Type III. In particular, $D = 0$ does not give a strict minimum.

This criterion does not depend on a finite search window. The condition $C > 0$ alone is insufficient for a barrier and minimum: the sign of B and the discriminant are essential.

4.2 Complete action

For S^3, $A = V^2$ and $C = 9\eta^2 - 36$. When $V \neq 0$,

$$|\eta| < 2 \quad \Longleftrightarrow \quad \text{type-II}, \qquad r_0 = \frac{\sqrt{12 - 3\eta^2}}{|V|}. \tag{M4-02}$$

Equivalently, $a_0 = r_0/2$ and $|\chi| < 1$. The condition is independent of θ and symmetric under $\eta \mapsto -\eta$.

For the isotropic or fixed-shape torus, $C = 9\eta^2 \geq 0$; the FULL potential is increasing, except that it is flat when both torsion parameters vanish. For Nil, $C = 36\eta^2 + 3 > 0$ and FULL is strictly increasing. There is no Type I region in FULL on these paths and no stabilization driven by the complete NY term.

4.3 Diagnostic actions

For S^3 with $V \neq 0$, $|\eta| < 2$ gives Type II for any mixed coefficient. At $|\eta| = 2$, Type II requires $V\eta\theta < 0$. For $|\eta| > 2$, Type I occurs precisely when

$$V\eta\theta < 0, \qquad 16\eta^2\kappa^4\theta^2 > 3(\eta^2 - 4). \tag{M4-03}$$

For the isotropic or fixed-shape torus, Type I requires

$$V\eta\theta < 0, \qquad \kappa^2|\theta| > \frac{\sqrt{3}}{4}. \tag{M4-04}$$

For Nil, the condition is

$$V\eta\theta < 0, \qquad (64\kappa^4\theta^2 - 12)\eta^2 > 1. \tag{M4-05}$$

Equality at the discriminant boundary gives a stationary inflection, not a strict minimum. These are conditions for TT and positive REE, not for FULL.

4.4 Degenerate parameters and numerical boundaries

At $V = 0$, this ansatz has $A = B = 0$, so the potential is linear. On S^3, it runs away for $|\eta| < 2$, is flat at $|\eta| = 2$, and increases for $|\eta| > 2$. On T^3, it is flat at $\eta = 0$ and otherwise increases. On Nil, it always increases. The generic quadratic case $A = 0$, $B > 0$, $C < 0$ would give Type II, but it is not the $V = 0$ specialization of this ansatz.

In floating-point evaluation, if $D \leq 64\epsilon_{\text{float}} \max(B^2, |3AC|)$, the implementation reports no resolved strict minimum. Exact $D = 0$ and the numerically unresolved neighborhood of that boundary are distinct: in particular, a small positive discriminant is not proved to be zero by this test. Analytic conditions take precedence. Appendix C distinguishes analytic classification from search-window and numerical-error behavior.

4.5 Figures and summary tables

See Table 3 for the conditions and values.

The construction, curves, or classification are illustrated in Figure 4, Figure 5, Figure 6, Figure 7.

Table 3: Analytic classification for $P, A > 0$, with $D = B^2 - 3AC$. For $A = 0$, classify the lower-degree polynomial directly.

Condition	Type	Extrema on positive scales
$C < 0$	II	Strict minimum
$C = 0,\ B < 0$	II	Minimum; the origin is excluded
$C > 0,\ B < 0,\ D > 0$	I	Maximum followed by a strict minimum
Otherwise	III	No strict minimum

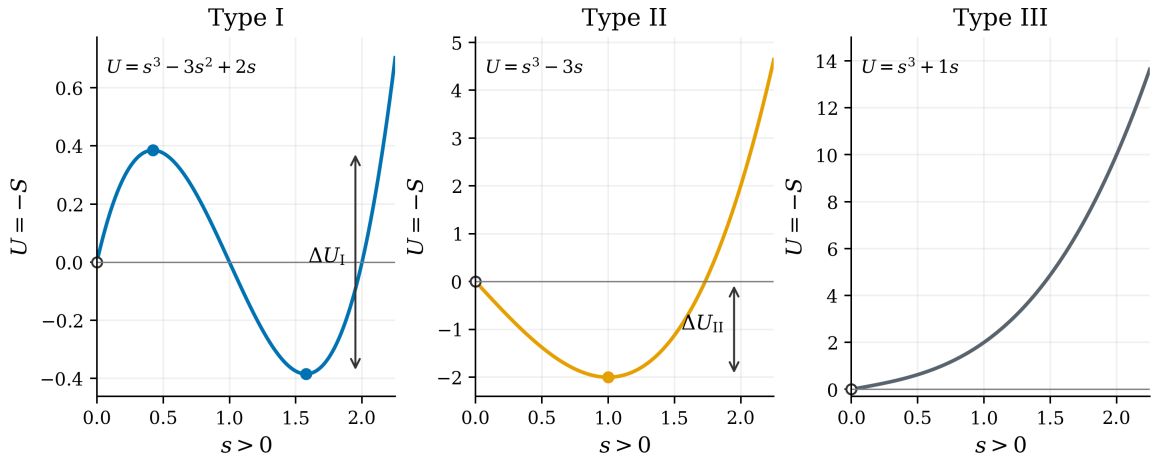


Figure 4: Schematic classification on positive scales, with $P = 1$ and each polynomial given in its panel. Open origin markers are excluded from the domain. Type I height is the maximum-to-minimum difference; Type II depth is measured from the origin limit. The Type III panel shows one example without a minimum.

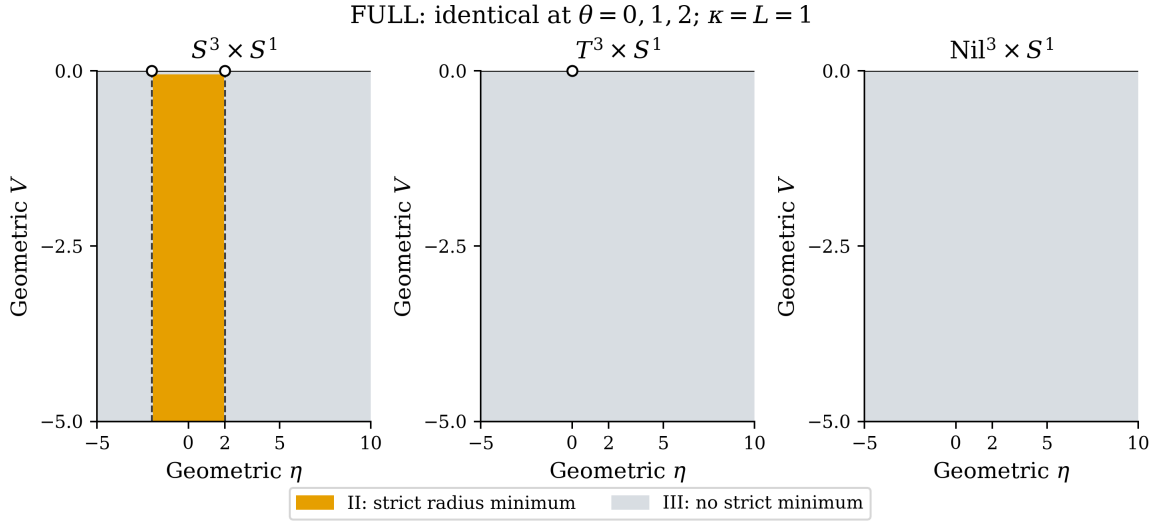


Figure 5: FULL phase maps at $\kappa = L = 1$, sampled at spacing 0.1 in geometric $V \in [-5, 0]$ and $\eta \in [-5, 10]$. Classification agrees at $\theta = 0, 1, 2$. Only S^3 with $V \neq 0$, $|\eta| < 2$ is Type II. Dashed boundaries $\eta = \pm 2$ and the top edge $V = 0$ are Type III; open circles mark flat cases. There is no Type I region.

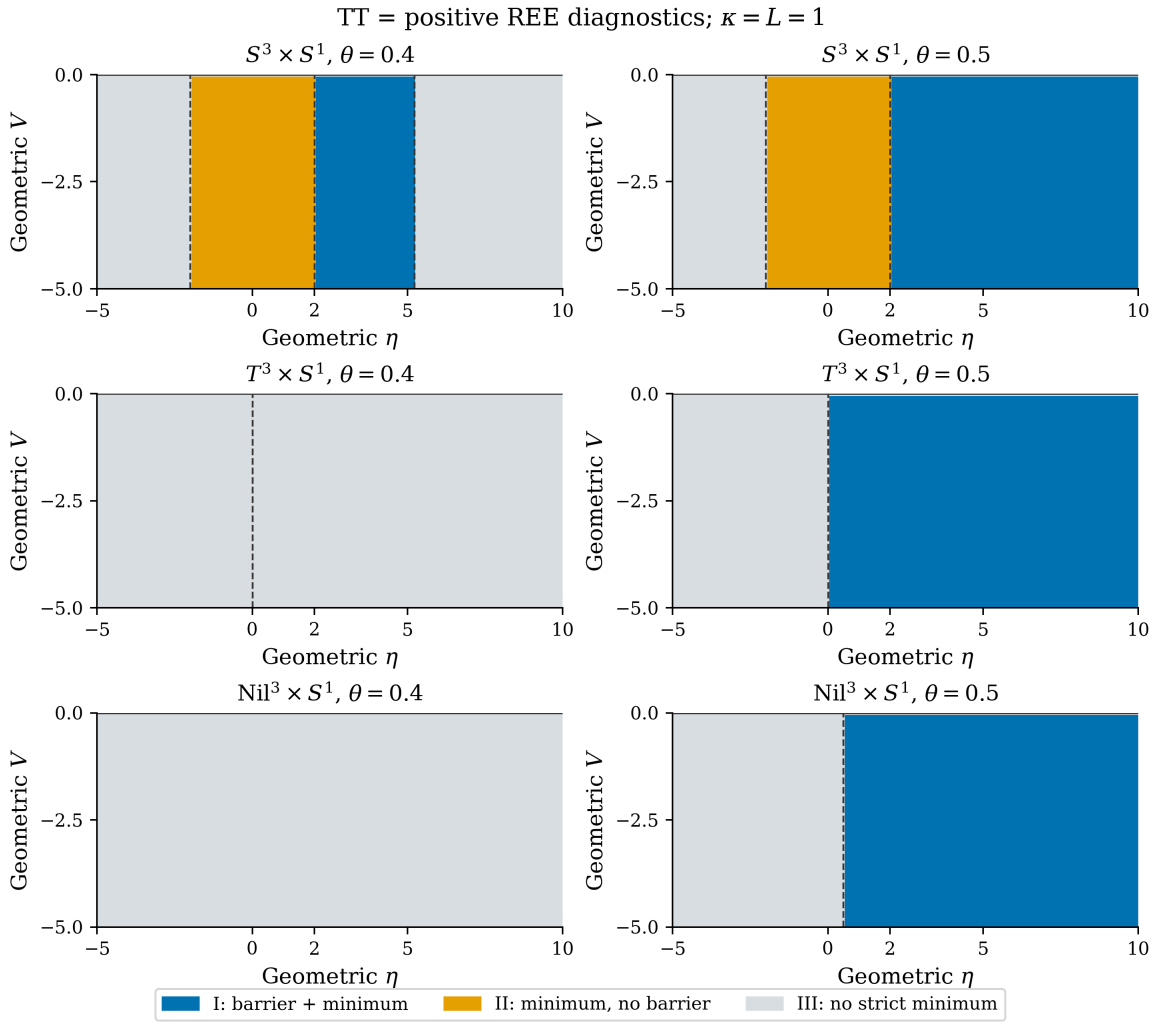


Figure 6: TT and positive REE phase maps for $\theta = 0.4, 0.5$, on the same input grid as Figure 5, with $\kappa = L = 1$. Dashed lines are the analytic boundaries of (M4-03)–(M4-05); colored cells have spacing 0.1. The top edge $V = 0$ is entirely Type III. For S^3 at $V < 0$, $\eta = 2$ belongs to Type II, whereas $\eta = -2$ is Type III. A positive repeated stationary root at $D = 0$ is Type III.

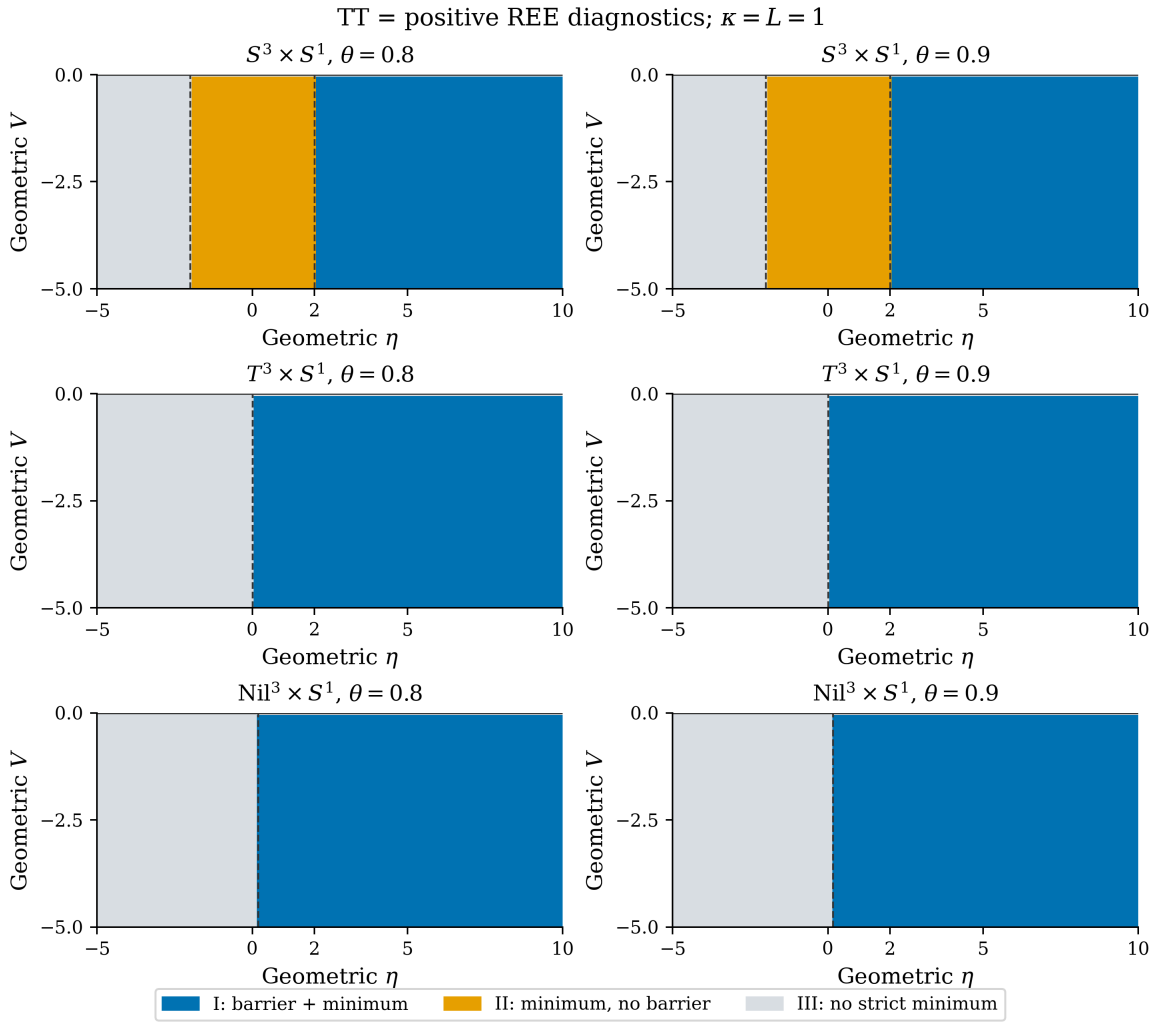


Figure 7: TT and positive REE phase maps for $\theta = 0.8, 0.9$. Domains, grid spacing, and boundary conventions are those of Figure 6. Type I regions of the diagnostic actions are not regions of the complete NY action.

5 Extremum values and their interpretation

5.1 Local curvature and potential height

At a strict cubic minimum $r_0 = r_+$,

$$U(r_0) = -Pr_0^2(2Ar_0 + B), \qquad U''(r_0) = 2P\sqrt{B^2 - 3AC}. \tag{M5-01}$$

The second derivative is the curvature of the one-variable reduced potential. It is neither a mass squared nor a statement that the full Hessian is positive.

For Type I, the height difference between the local maximum and minimum is

$$\Delta U_{\mathrm{I}} = U(r_-) - U(r_+) = \frac{4P(B^2 - 3AC)^{3/2}}{27A^2} > 0. \tag{M5-02}$$

For Type II, we instead use the depth relative to the algebraic value $U(0) = 0$:

$$\Delta U_{\mathrm{II}} = -U(r_+) = Pr_+^2(2Ar_+ + B) > 0. \tag{M5-03}$$

The point $r = 0$ is a degenerate geometry outside the domain. This depth is not a barrier between two positive-radius extrema.

5.2 The FULL minimum on the sphere

For $V \neq 0$ and $|\eta| < 2$,

$$\begin{aligned} r_0 &= \frac{\sqrt{12 - 3\eta^2}}{|V|}, \\ U(r_0) &= -\frac{\pi^2 L}{6\kappa^2}\frac{(12 - 3\eta^2)^{3/2}}{|V|}, \\ U''(r_0) &= \frac{\pi^2 L}{2\kappa^2}|V|\sqrt{12 - 3\eta^2}. \end{aligned} \tag{M5-04}$$

The negative linear term reflects the competition between positive spatial curvature and axial torsion, while the positive cubic term comes from vector torsion. The NY coupling does not enter this mechanism. As $V \to 0$, the radius tends to infinity and the finite minimum is lost to the runaway limit. As $|\eta| \to 2$, the radius approaches the excluded origin. With zero torsion, $U = -3\pi^2 Lr/\kappa^2$, so there is no finite well. Curvature alone does not create this minimum.

5.3 Distinct boundaries

In the region $C > 0$, $B < 0$, the condition that the diagnostic minimum have the same height as the algebraic origin is

$$B^2 = 4AC, \qquad r_+ = -\frac{B}{2A}. \tag{M5-05}$$

This is distinct from the saddle-node condition $B^2 = 3AC$. At the saddle-node the strict minimum is lost, whereas at $B^2 = 4AC$ it remains and crosses the reference height. The Type I/II distinction is determined by the presence of a positive-radius maximum, not by the minimum's height relative to the origin.

5.4 Signs, normalization, and comparisons

FULL depends on V^2 and η^2. The diagnostic mixed term depends on $V\eta\theta$ and is invariant under simultaneous reversal of V and η, as in the T/τ dictionary. Reversing only η generally changes the diagnostic potential.

A positive prefactor preserves radii and phase types but scales all potential heights. Comparing different geometries requires specifying the global quotient, units, and physical quantities held fixed.

6 Representative parameters and scaling

6.1 A FULL sphere minimum

Set $\kappa = L = 1$ and use geometric-torsion parameters throughout. For $V = -4$, $\eta = 0$, and arbitrary θ,

$$r_0 = \frac{\sqrt{3}}{2}, \qquad a_0 = \frac{\sqrt{3}}{4}, \qquad U(r_0) = -\sqrt{3}\,\pi^2, \qquad \Delta U_{\mathrm{II}} = \sqrt{3}\,\pi^2. \tag{M6-01}$$

This is a Type II minimum. Its positive depth is measured relative to the algebraic origin as in Section 5. At $\eta = \pm 2$, the FULL potential has only its positive cubic term. At $\eta = \pm 3$, its linear term is also positive, and there is no strict positive-radius minimum. FULL does not select one sign of η.

6.2 Torus and Nil examples

For the isotropic torus with $V = -4$, $\eta = 0$, $\mathcal{R}_{\mathrm{EC}} = -32/3$ and $U = 128\pi^3 r^3/3$. The vanishing Levi–Civita curvature does not imply vanishing torsionful curvature or a flat potential. The FULL potential on Nil is increasing for every real η, V.

For $V = -4$, $\eta = 3$, and $\kappa^2\theta = 1$, the diagnostic action has Type I behavior on all three geometries, while FULL has no strict minimum on any of the three paths. At $\eta = 0$ or $V = 0$, the mixed density vanishes and all variants coincide. This does not imply equality of FULL and REE for general MX inputs.

6.3 Scaling with vector torsion

For $V \neq 0$, define $x = |V|r$ and $\sigma = \operatorname{sign} V$. The diagnostic sphere potential becomes

$$U_{\mathrm{TT}}^{S^3} = \frac{\pi^2 L}{12\kappa^2|V|}\left[x^3 + 12\sigma\eta\kappa^2\theta x^2 + (9\eta^2 - 36)x\right]. \tag{M6-02}$$

At fixed $\eta, \kappa^2\theta, \sigma$ (and fixed κ, L), extremum radii and potential-height differences scale as $1/|V|$, provided the corresponding minimum exists. The same substitution applies to the isotropic torus and Nil potentials. For example, there is no FULL minimum at $\eta = 3$ to which a minimum scaling law could be applied. The case $V = 0$ is a separate linear problem.

6.4 Figures and summary tables

See Table 4 for the conditions and values.

The construction, curves, or classification are illustrated in Figure 8, Figure 9.

Table 4: Representative analytic values at $V = -4, \theta = \kappa = L = 1$, with $s_0 = r_0$ or R_0. TT equals positive REE. ΔU denotes a Type I barrier or a Type II depth from the origin limit. Dashes indicate a missing minimum.

Space	Action	η	Type	s_0	$U_{\min}$	ΔU
S^3	FULL	3	III	—	—	—
S^3	FULL	2	III	—	—	—
S^3	FULL	0	II	0.866025	−17.0947	17.0947
S^3	FULL	−3	III	—	—	—
S^3	TT	3	I	5.83945	−1202.1	1205.04
T^3	FULL	3	III	—	—	—
T^3	FULL	2	III	—	—	—
T^3	FULL	0	III	—	—	—
T^3	FULL	−3	III	—	—	—
T^3	TT	3	I	5.70416	−103666	104640
Nil^3	FULL	3	III	—	—	—
Nil^3	FULL	2	III	—	—	—
Nil^3	FULL	0	III	—	—	—
Nil^3	FULL	−3	III	—	—	—
Nil^3	TT	3	I	5.70127	−649133	655365

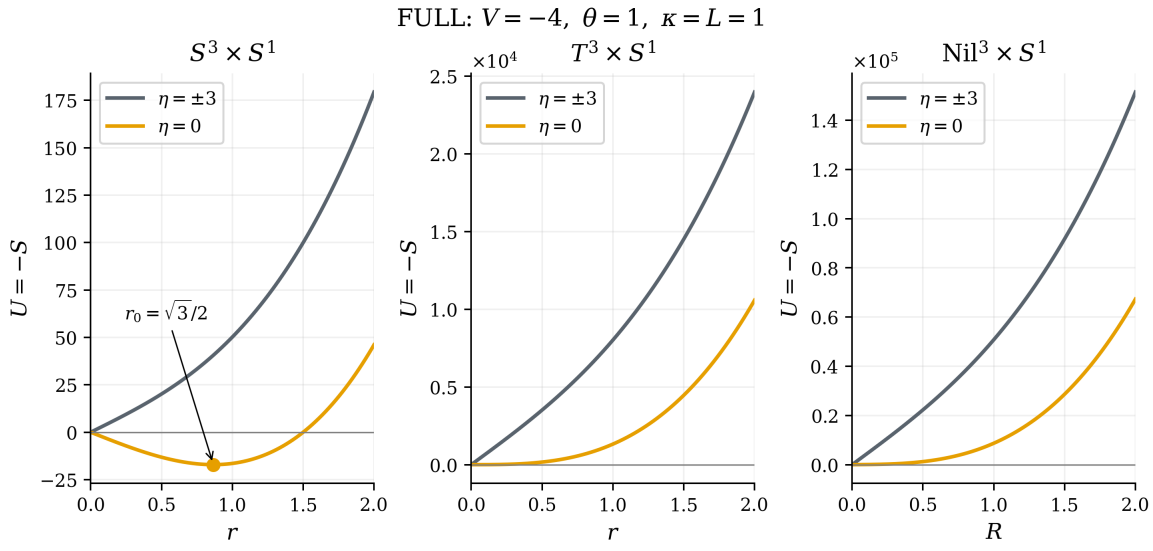


Figure 8: Representative FULL curves at $V = -4, \theta = \kappa = L = 1$. Curves for $\eta = 3$ and -3 coincide. Only the sphere at $\eta = 0$ has a strict positive-radius minimum, at $r_0 = \sqrt{3}/2$ (round radius $a_0 = \sqrt{3}/4$). Numerical values are given in Table 4.

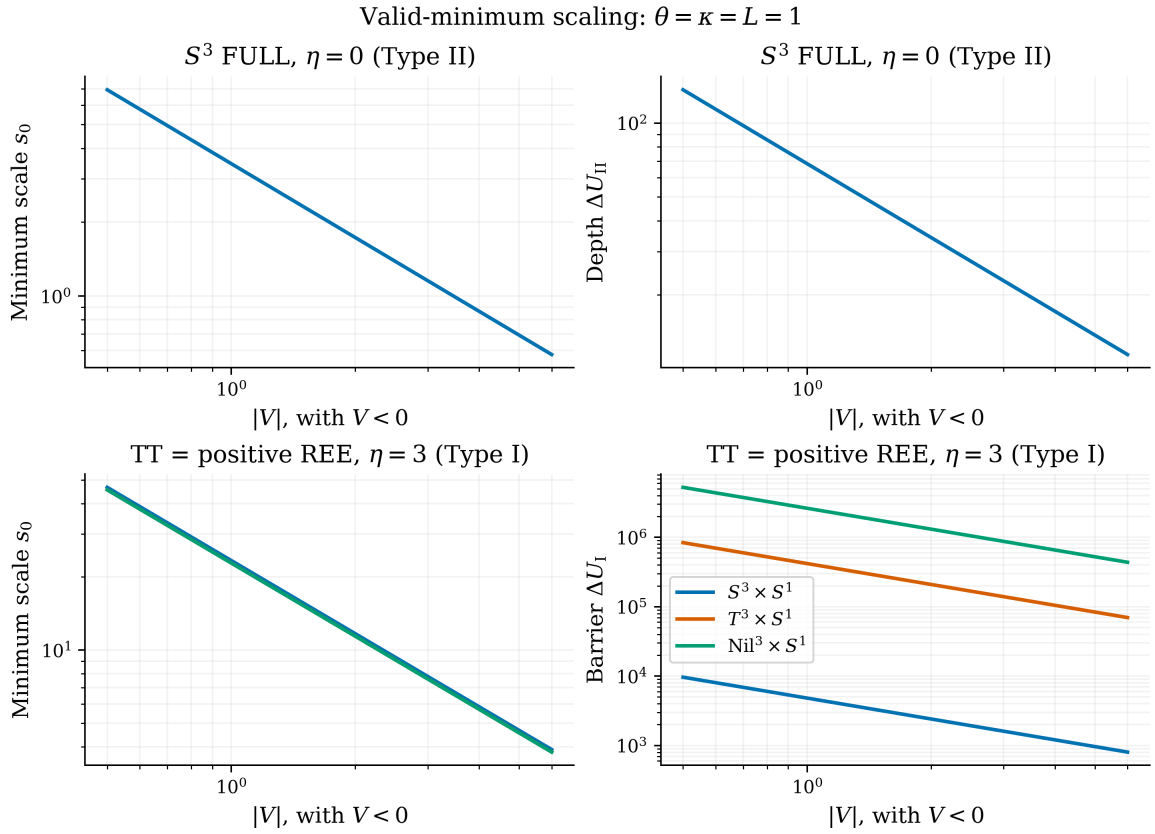


Figure 9: Scaling where minima exist, at $\theta = \kappa = L = 1$ and 121 values of $|V| \in [0.5, 6]$ with $V < 0$. Top: the sphere FULL minimum and Type II depth at $\eta = 0$. Bottom: diagnostic Type I minima and barriers at $\eta = 3$ for all geometries. Each scales as $1/|V|$. Here $s_0 = r_0$ or R_0; bottom-left curves nearly coincide. FULL has no minimum at $\eta = 3$.

7 Variation scope and physical interpretation

7.1 A radius minimum is not full stationarity

On S^3 in FULL, take $V = 1$, $\eta = \theta = 0$, and $\kappa = L = 1$. At $r = 2\sqrt{3}$,

$$\begin{aligned} \partial_r U &= 0, & \partial_r^2 U &= \sqrt{3}\,\pi^2 > 0, \\ \partial_V U &= 4\sqrt{3}\,\pi^2 \neq 0, & \partial_L U &= -4\sqrt{3}\,\pi^2 \neq 0. \end{aligned} \tag{M7-01}$$

Thus radius stationarity and positive radial curvature coexist with nonzero derivatives in other directions. On T^3, fixed-shape scaling and variation of the three radii independently are different questions. In particular, Appendix E shows that the full radius Hessian has $H_{22} = H_{33} = 0$ on the specified family.

7.2 Scoped nonexistence of a full vacuum on the sphere product

The following statement concerns standard EC gravity with a constant complete NY coupling, no sources, and no cosmological constant, on a closed real Euclidean $S^3 \times S^1$. The coframe is smooth and nondegenerate and the metric-compatible connection is varied independently.

The complete NY term is a total derivative, so a constant coefficient does not change the bulk equations on a closed manifold. For a nondegenerate coframe, the standard EC connection

equation gives zero torsion, and the coframe equation gives Ricci flatness. For these bulk variations and the NY identity, see [4, Sec. IV.A–B, Eqs. (4.10)–(4.12), (4.20)]. We now apply their geometric consequences under the real Euclidean, closed-manifold assumptions stated above.

Since $\chi(S^3 \times S^1) = 0$, the four-dimensional Gauss–Bonnet formula for a Ricci-flat Riemannian metric gives zero for a positive constant times the integral of the squared Riemann-curvature norm. The nonnegative integrand must therefore vanish everywhere, so the metric would be flat.

By Bieberbach's theorem, a compact flat four-manifold has a finite cover by a flat T^4; see also the proof of Lemma 1 in [5]. Its fundamental group must contain a finite-index subgroup isomorphic to $\mathbb{Z}^4$. But $\pi_1(S^3 \times S^1) = \mathbb{Z}$, whose subgroups are cyclic and cannot be isomorphic to $\mathbb{Z}^4$. This contradiction excludes a full vacuum satisfying all the stated assumptions. It is a global obstruction, not an inference from a finite set of parameter examples. It does not exclude the off-shell geometries used in this paper, and does not address theories with a cosmological constant, sources, boundaries, complex metrics, nonconstant θ, or higher-curvature terms.

7.3 NY, curvature contractions, and topology

The contraction $\varepsilon^{abcd} R_{abcd}$ is linear in curvature. It is not the Pontryagin density, which is quadratic in curvature through $R \wedge R$. Its vanishing on a torsion-free slice does not imply vanishing for general torsion. Likewise, a zero integral of the complete NY form does not require the separate TT and REE integrals to vanish.

7.4 What is required for physical observables

The sign of $U_{\min}$ alone does not identify a dominant saddle in a path integral. Such an interpretation requires an integration contour, a weighting prescription, and the relevant variations. Nor does a height difference ΔU give a physical energy or a tunneling rate without a kinetic term, constraints, boundary conditions, and a bounce or an equivalent dynamical analysis. The Type I action difference alone does not determine a WKB exponent. Fixing the global volume does not supply that physical unit dictionary.

Comparing topology sectors requires a common physical framework, including the quantities and constraints held fixed. An extension to $S^2 \times \mathbb{R}$ would also require a noncompact-volume prescription and a defined reduction. Nontriviality of TS^2 alone does not prove the absence of a global frame on a product containing it. No such extension is computed here.

8 Conclusions and open questions

For the static homogeneous axial-plus-vector ansatz on $S^3 \times S^1$, $T^3 \times S^1$, and compact $\mathrm{Nil}^3 \times S^1$, the TT and positive-REE densities coincide pointwise. The complete NY density vanishes, so the FULL reduced potential is independent of its constant coupling θ.

At fixed $\eta, V, L, \kappa, \theta$, FULL has a strict sphere radius minimum precisely when $V \neq 0$ and $|\eta| < 2$; this is Type II. It has no strict minimum along the isotropic or fixed-shape torus path or the uniform Nil path. The torus path with two radii fixed is a different restriction and can have a radius minimum. None of these radius statements establishes stability under all variations.

The diagnostic TT and positive-REE potentials coincide and can support Type I barriers under the conditions in Section 4. The cubic analysis distinguishes strict minima, boundary roots, stationary inflections, and degenerate linear cases without relying on a finite scan. These diagnostic barriers must not be attributed to the complete NY action.

The normalization is part of each geometric definition: $a = r/2$ for the sphere, spatial periods 2π for the torus, the Nil lattice $(2\pi, 2\pi, 4\pi^2)$, and the actual circle circumference L. The geometric torsion T and code input $\tau = -T$ describe the same connection after simultaneous parameter conversion.

The results provide a method for comparing geometric reductions. They do not identify a full vacuum, a particle, or a topology selected by physical dynamics. Under the restricted source-free, constant-NY, real Euclidean assumptions of Section 7, a smooth nondegenerate full vacuum on $S^3 \times S^1$ is excluded; the off-shell reductions remain well defined.

Open physical questions include the definition of energy, mass, and decay observables, and the common constraints needed to compare different topology sectors. These require additional dynamical and physical input beyond the radius potentials established here.

A Connection components and the code-input dictionary

A.1 Levi–Civita connection

Indices and structure coefficients follow Section 2. The nonzero components on S^3 are

$$\Gamma^{\mathrm{LC}}_{012} = -\Gamma^{\mathrm{LC}}_{021} = \Gamma^{\mathrm{LC}}_{120} = -\Gamma^{\mathrm{LC}}_{102} = \Gamma^{\mathrm{LC}}_{201} = -\Gamma^{\mathrm{LC}}_{210} = \frac{2}{r}. \tag{A-01}$$

All components vanish on T^3. For Nil, the nonzero components are

$$\Gamma^{\mathrm{LC}}_{012} = \Gamma^{\mathrm{LC}}_{021} = \frac{1}{2R}, \quad \Gamma^{\mathrm{LC}}_{102} = \Gamma^{\mathrm{LC}}_{120} = -\frac{1}{2R}, \quad \Gamma^{\mathrm{LC}}_{201} = -\Gamma^{\mathrm{LC}}_{210} = -\frac{1}{2R}. \tag{A-02}$$

They satisfy $C + \Gamma^{\mathrm{LC}}_{\cdot cb} - \Gamma^{\mathrm{LC}}_{\cdot bc} = 0$. The spatial scale is constant with respect to frame position.

A.2 Contortion from geometric torsion

Substituting the axial and vector parts in (M2-03) gives

$$K^{\mathrm{ax}}_{ijk}[T] = -\frac{\eta}{s}\varepsilon_{ijk}, \qquad K^{\mathrm{vec}}_{abc}[T] = -\frac{1}{3}(\delta_{ac}V_b - \delta_{bc}V_a). \tag{A-03}$$

Cyclic permutations of the axial component are even, so the three-term combination in the contortion formula reduces to T_{ijk}. For the vector component, the $\delta_{ab}V_c$ terms cancel. Both contortion components are antisymmetric in their first two indices.

A.3 The input array

Denote the torsion input array of the accompanying code by τ_{abc}. The code constructs the connection using

$$K^{\mathrm{code}}_{abc}[\tau] = \frac{1}{2}(\tau_{abc} + \tau_{bca} - \tau_{cab}), \qquad \Gamma^{\mathrm{code}} = \Gamma^{\mathrm{LC}} + K^{\mathrm{code}}. \tag{A-04}$$

The geometric torsion reconstructed from this connection by the Cartan equation is $T = -\tau$. The same-connection correspondence is therefore

$$\tau = -T, \qquad \eta_\tau = -\eta, \qquad V_\tau = -V, \qquad K^{\mathrm{code}}[-T] = K[T]. \tag{A-05}$$

Simultaneously converting the corresponding parameters gives the same connection, curvature, and action in both descriptions. This does not mean reversing K alone, or reversing η while leaving V fixed.

In code-input variables, axial contortion is $+\eta_\tau\varepsilon_{ijk}/s$, and vector contortion is $+(\delta_{ac}(V_\tau)_b - \delta_{bc}(V_\tau)_a)/3$. Both axial and vector geometric-torsion components have the opposite sign, while $\eta^2, V^2, V\eta$ are unchanged.

A.4 Scales and input domains

An input interval $\eta_\tau \in [p, q]$ maps to $\eta \in [-q, -p]$, and $V_\tau \in [v_-, v_+]$ maps to $V \in [-v_+, -v_-]$. When plotting axes in ascending order, the associated data order must also be reversed.

For the actual sphere radius,

$$a = \frac{r}{2}, \qquad \frac{2\eta}{r} = \frac{2\chi}{a}, \qquad \chi = \frac{\eta}{2} = -\frac{\eta_\tau}{2}. \tag{A-06}$$

For torus circumferences,

$$\ell_i = 2\pi R_i, \qquad \frac{2\eta}{R_1} = \frac{2\zeta}{\ell_1}, \qquad \zeta = 2\pi\eta = -2\pi\eta_\tau. \tag{A-07}$$

For Nil, the adopted lattice is fixed and R is the common scale. In all three geometries, L is the actual circle circumference and the coordinate u has period one.

A.5 Independent ingredients in action comparisons

The simultaneous sign conversion is a change of description of the same connection. The two-form coefficients in NY, the integration volume, and the action variant are separate definitions; they are not absorbed by the sign dictionary. Both descriptions use the same $n_{\mathrm{REE}} = \varepsilon R/2$ and the adopted volumes.

B Checks of the connection, invariants, and classification

B.1 Objects to be checked

The symbolic calculation constructs the Levi–Civita connection from the structure coefficients, adds the torsion ansatz, and computes curvature and contractions. Verification compares exterior-form and component calculations, rather than merely redisplaying the same expressions. The scope is three geometries, three torsion modes (AX, VT, MX), and three action variants (FULL, TT, REE).

Basic conditions are $C^a{}_{bc} = -C^a{}_{cb}$, $d^2e^a = 0$, torsion reconstruction by the Cartan equation, and metric compatibility. With the last connection index denoting differentiation direction,

$$\nabla_c \delta_{ab} = -\Gamma^d{}_{ac}\delta_{db} - \Gamma^d{}_{bc}\delta_{ad} = 0. \tag{B-01}$$

This is equivalent to $\Gamma_{abc} + \Gamma_{bac} = 0$. The geometric torsion represented by an input array is determined separately by the Cartan equation.

B.2 Curvature and identities

Metric-compatible curvature is antisymmetric in the first index pair, and the two-form definition gives antisymmetry in the last pair. Pair exchange $R_{abcd} = R_{cdab}$ is not required for a general connection with torsion. For example, the MX ansatz can give

$$R_{0123} - R_{2301} = \frac{2V\eta}{3s}. \tag{B-02}$$

This supplies a counterexample to treating pair exchange as a universal test. The first and second Bianchi identities are checked in their torsionful exterior-form expressions,

$$DT^a = R^a{}_b \wedge e^b, \qquad DR^a{}_b = 0. \tag{B-03}$$

For NY, construct $d(e^a \wedge T_a)$, $T^a \wedge T_a$, and $e^a \wedge e^b \wedge R_{ab}$ independently and compare the coefficients and difference in (M2-06). The cuts $V = 0$, $\eta = 0$, and $\theta = 0$ also check the agreement of action variants and the EC part.

B.3 Symbolic verdicts and numerical comparisons

The engine distinguishes `PROVED_ZERO`, where zero is proved; `WITNESS_NONZERO`, where a nonzero witness is found; and `UNPROVED`, where neither is established. An unproved expression is not treated as zero, and a concrete nonzero example is distinguished from proof of a symbolic identity.

The reduced potentials can be checked by independently calculating curvature and volume, integrating, and comparing with Section 3. Further independent calculations cover first and second derivatives, discriminants, positive stationary roots, and action differences between maxima and minima. Representative cases include minima near the origin, large-radius minima, $C = 0$, $D = 0$, $V = 0$, and both torsion coefficients zero.

Agreement at finitely many numerical points does not replace an identity. The analytic formulas are checked symbolically, while independent multiprecision evaluation checks classifier roots, heights, and near-boundary behavior. Comparisons require corresponding T and τ parameters, the adopted volume, and the same variation path.

B.4 Reproducing calculations

Record the code version, inputs, logs, and outputs together. For each formula comparison, specify the geometry, action variant, torsion mode, and identity tested. Appendix F gives execution procedures; the component conventions provide the code's geometric definitions.

B.5 Figures and summary tables

See Table 5 for the conditions and values.

Table 5: Local invariants at $s = 1, V = -4, \eta = 3$. Here $\mathcal{T}^2 = T_{abc}T^{abc}$; n is the density relative to the volume form, and $n_{\mathrm{FULL}} = n_{\mathrm{TT}} - n_{\mathrm{REE}}$.

Space	$\mathcal{R}_{\mathrm{LC}}$	$\mathcal{R}_{\mathrm{EC}}$	$\mathcal{T}^2$	$n_{\mathrm{TT}} = n_{\mathrm{REE}}$	n_{FULL}
S^3	24	$-122/3$	$680/3$	48	0
T^3	0	$-194/3$	$680/3$	48	0
Nil^3	$-1/2$	$-391/6$	$680/3$	48	0

C Numerical classification and input domains

C.1 Input meaning and ranges

In the accompanying parameter-scan script, arguments `V` and `eta` are the coefficients V_τ, η_τ of the array defined in Appendix A. Convert to main-text geometric torsion by $V = -V_\tau$, $\eta = -\eta_\tau$; θ, the scales, κ, and L are unchanged.

The CLI defaults are $V_\tau \in [0, 5]$, $\eta_\tau \in [-10, 5]$, and $\theta \in [0, 5]$, corresponding to $V \in [-5, 0]$, $\eta \in [-5, 10]$. Endpoints are included; $V = 0$ is not automatically excluded. The respective point

counts are 51, 151, and 51, giving 392,751 inputs per geometry and variant, or 3,534,759 inputs for all three geometries and all three variants. These numbers describe the size of an input grid.

The Python function `run_scan` has different defaults: 6, 11, and 6 points, with $\eta_\tau \in [-5, 5]$. A record of default settings must identify the entry point. The interactive viewer's default domains are $V_\tau \in [0, 10]$, $\eta_\tau \in [-15, 5]$, $\theta \in [0, 10]$, corresponding to $V \in [-10, 0]$, $\eta \in [-5, 15]$. These are subsets of the analytic parameter domain, not restrictions on the formulas themselves.

The CLI interval becomes $\chi \in [-5/2, 5]$ when using the actual sphere radius, and $\zeta \in [-10\pi, 20\pi]$ when using torus circumferences. For example, the input pairs (V_τ, η_τ) $(4, -3)$, $(4, -2)$, $(4, 0)$, and $(4, 3)$ correspond respectively to main-text pairs $(-4, 3)$, $(-4, 2)$, $(-4, 0)$, and $(-4, -3)$.

C.2 The cubic classifier

For registered potentials and callables supporting coefficient retrieval, `get_cubic_coefficients` returns the coefficients of $U = ar^3 + br^2 + cr$, and `analyze_cubic_stability` classifies them. Here $(a, b, c) = P(A, B, C)$ includes the positive prefactor. Signs, discriminants, and roots are treated as in Section 4.

Coefficient and root arithmetic uses Decimal with 50-digit precision. This does not give 50-digit accuracy to floating-point coefficients supplied as input. To control cancellation, for $b \geq 0, c < 0$ the positive root is evaluated as $-c/(b + \sqrt{b^2 - 3ac})$.

The unresolved discriminant band is set by $64\epsilon_{\text{float}} \max(b^2, |3ac|)$. A Type III return in this band means that no strict minimum was resolved; it does not prove that a small positive discriminant is exactly zero. Boundary decisions should include the analytic inequalities.

If coefficients cannot be obtained for a general callable, a separate bounded numerical search is used as a fallback. Its conclusions are limited by the search window and resolution, and do not have the same guarantees as the registered cubic criterion. Exceptions from nonfinite coefficients or invalid inputs are not counted as determinations of a new physical phase.

C.3 Output format and preservation

The CSV has eight columns:

```
topology, ny_variant, V, eta, theta, r0, delta_V, stability_type
```

The columns `V`, `eta` retain V_τ, η_τ. For S^3, `r0` is the code scale r, not the actual radius a. For Type I, `delta_V` is the difference from the positive-radius maximum to the minimum; for Type II, it is the depth from the origin limit. For Type III without a minimum, `r0` and `delta_V` are blank. These two kinds of height must not be plotted as the same kind of barrier.

The aggregate filename is `dppu_scan_all_{timestamp}.csv`, and individual filenames are `dppu_scan_{topology}_{ny_variant}_{timestamp}.csv`. Since the CSV does not contain all code-version and volume-convention information, preserve the command, version, conventions, and log separately.

C.4 Interpreting aggregate results

Phase fractions and failure rates require an input population for which the calculation actually completed. Record unresolved numerical boundaries, input errors, failed numerical searches, and

exactly monotone functions as distinct cases. A scan classifies the selected grid; conditions over the whole parameter domain follow from the analytic results of Section 4.

C.5 Figures and summary tables

See Table 6 for the conditions and values.

Table 6: Input-to-text dictionary. Apply $V = -V_\tau, \eta = -\eta_\tau$ simultaneously, leaving θ, κ, L and coframe scales unchanged. Physical length conversions follow Appendix A.

Input / quantity	τ / coframe	T / geometric display
CLI	$V_\tau \in [0,5],\ \eta_\tau \in [-10,5]$	$V \in [-5,0],\ \eta \in [-5,10]$
run_scan	$V_\tau \in [0,5],\ \eta_\tau \in [-5,5]$	$V \in [-5,0],\ \eta \in [-5,5]$
Viewer	$V_\tau \in [0,10],\ \eta_\tau \in [-15,5]$	$V \in [-10,0],\ \eta \in [-5,15]$
S^3	$r,\ \eta_\tau$	$a = r/2,\ \chi = -\eta_\tau/2$
T^3	$R_i,\ \eta_\tau$	$\ell_i = 2\pi R_i,\ \zeta = -2\pi\eta_\tau$

D Displaying potentials and phase diagrams

D.1 Coordinates, parameters, and plotted quantities

Potential curves and phase diagrams specify the geometry, action variant, variation path, and fixed κ, L and torsion coefficients. Conversion from code parameters V_τ, η_τ to the main-text V, η reverses both signs according to Appendix A. Ordering an axis increasingly also requires reordering the associated data.

Use $a = r/2$ for the actual sphere radius and $\ell_i = 2\pi R_i$ for torus circumferences. Convert the axial coefficients as well: $\chi = \eta/2$ and $\zeta = 2\pi\eta$, respectively.

D.2 Extrema and phase labels

For Type I, ΔU is the action difference between a positive-radius maximum and minimum; Type II depth uses the origin limit as its reference. The displayed quantity must identify this distinction. A radius minimum does not establish a vacuum of the full field equations or dynamical stability.

Type III includes monotone functions, flat functions, runaway behavior, and stationary inflections. An unresolved discriminant band or a failed calculation is recorded separately from an analytic classification. A missing minimum radius in a CSV is not displayed as a minimum at zero.

D.3 Matching curves to numerical outputs

Relating a phase label or extremum to a curve requires the same potential, input parameters, and variation path. A positive overall volume factor preserves extremum positions and types, and scales action values and heights. Comparisons of vertical scales across geometries use the volumes and length conventions of Section 3. Appendix C defines numerical columns; Appendix F gives calculation procedures.

D.4 Data accompanying the figures

The repository directory `data/paper_v3` provides input domains, data columns, and the PNG/SVG index for Figures 1–9 and Tables 1–7. These data use geometric V, η and a distinct schema from the scanner CSV in Appendix C.

E Anisotropic torus and variation paths

E.1 General reduced function

All spatial angles have period 2π, and the circle coordinate has period one. For $R_i > 0$, define the axial torsion as $2\eta/R_1$. This singles out R_1, so simply permuting the R_i need not give the same variation problem:

$$\begin{aligned} U_{\mathrm{FULL}}(R_1, R_2, R_3) &= QR_2R_3\left(V^2R_1 + \frac{9\eta^2}{R_1}\right), \\ U_{\mathrm{TT}} = U_{\mathrm{REE}} &= QR_2R_3\left(V^2R_1 + \frac{9\eta^2}{R_1} + 12V\eta\kappa^2\theta\right), \\ Q &= \frac{8\pi^3 L}{3\kappa^2} > 0 \end{aligned} \tag{E-01}$$

The quantities $\eta, V, L, \kappa, \theta$ are held fixed.

E.2 Fixed shape and isotropic expansion

For fixed $\alpha, \beta > 0$, with $R_2 = \alpha R_1$, $R_3 = \beta R_1$, the isotropic potential in Section 3 is multiplied by $\alpha\beta$. Radius-extremum positions and types are unchanged, while action values and heights acquire that factor. There is no strict minimum in FULL along this path.

This equivalence concerns expansion with shape held fixed; it does not imply stability in the α, β directions. Other constraints, such as volume-preserving shape variations, require their own reduced functions.

E.3 Varying one radius with two radii fixed

If R_2, R_3 are fixed and only R_1 varies, $V\eta \neq 0$ gives a strict minimum:

$$R_{1,0} = 3\left|\frac{\eta}{V}\right|, \qquad \frac{\partial^2 U}{\partial R_1^2} = QR_2R_3\frac{18\eta^2}{R_1^3} > 0. \tag{E-02}$$

The mixed term of the diagnostic actions is constant on this path, so the minimum position and second derivative agree with FULL. However, for $\eta \neq 0$, $U \to +\infty$ as $R_1 \to 0$. The Type II cubic depth formula, which uses a zero origin limit, does not apply to this minimum.

For $\eta = 0, V \neq 0$ the potential increases; for $V = 0, \eta \neq 0$ it decreases from positive values toward zero; when both vanish it is flat. These cases are distinguished from isotropic expansion.

E.4 Independent variations of all three radii

Write $U = R_2 R_3 f(R_1)$. Then

$$\begin{aligned} \partial_{R_1} U = R_2 R_3 f', \quad \partial_{R_2} U = R_3 f, \quad \partial_{R_3} U = R_2 f, \\ H_{22} = H_{33} = 0, \qquad H_{23} = f. \end{aligned} \tag{E-03}$$

A minimum in r or R_1 alone does not imply that the full gradient vanishes. Since $H_{22} = H_{33} = 0$, the Hessian in the three independent variables cannot be positive definite. In FULL, $f > 0$ at the R_1 minimum with $V\eta \neq 0$, so the other two directions are not stationary.

Even if a diagnostic action satisfies the special conditions $f = f' = 0$, flat directions in R_2, R_3 remain. Such a point is not a strict minimum in all variables. With additional physical constraints on shape variations, stability must be defined separately on the allowed subspace.

E.5 Figures and summary tables

See Table 7 for the conditions and values.

Table 7: Torus variation paths with $U = R_2 R_3 f(R_1)$ and fixed parameters. The constant diagnostic shift leaves the minimum position unchanged when two radii are fixed.

Variation path	FULL	TT / positive REE
$R_2 = \alpha R_1,\ R_3 = \beta R_1$	No strict minimum	Same type as the isotropic path
Fixed R_2, R_3	$R_{1,0} = 3\lvert\eta/V\rvert$ $(V\eta \neq 0)$	Same minimum position as FULL
Three independent radii	No nontrivial fully stationary point	Even if $f = f' = 0$, flat directions remain

F Code, data, and reproducibility

F.1 Access to code and materials

Code and materials are provided in Muacca/DPPUv2-paper01. The repository carries DOI 10.5281/zenodo.18213677. Citations should identify the version and files used.

The symbolic core is `script/DPPUv2_engine_core_v3.py`. Geometry-specific entry points are `script/DPPUv2_runner_S3S1_v3.py`, `script/DPPUv2_runner_T3S1_v3.py`, and `script/DPPUv2_runner_Nil3S1_v3.py`. The script `script/DPPUv2_parameter_scan_v3.py` evaluates the reduced potentials. The script guide and component and global conventions specify usage and definitions. Dependencies are listed in `script/requirements.txt`.

F.2 Environment and calculation examples

The following commands assume that the current directory is the repository root. To create a virtual environment, run `python-mvenv.venv` and activate it in the shell being used. Install dependencies with `python-mpipinstall-rscript/requirements.txt`.

An example symbolic calculation is:

```
python -B script/DPPUv2_runner_S3S1_v3.py --mode MX --ny-variant FULL --log-file new_s3_full.log
```

A small scan is:

```
python -B script/DPPUv2_parameter_scan_v3.py --topologies S3 --ny-variants FULL --V-min 1
    --V-max 4 --V-points 2 --eta-min -3 --eta-max 3 --eta-points 7 --theta-min 0 --theta-max 1
    --theta-points 2 --workers 1 --output-dir new_scan
```

These inputs have $V_\tau > 0$, corresponding to main-text $V < 0$. Use separate log and output destinations for each run, and retain the association between inputs and results.

F.3 Recording calculation conditions

Record the code version, action variant, T/τ input convention, volume convention, variation path, point counts and endpoints, and classifier boundary treatment. Distinguish r from the actual sphere radius a, and R_i from the torus circumferences ℓ_i. Specify the Nil geometry including its lattice face identifications.

Appendix C defines the eight CSV columns and input domains; Appendix D defines displayed quantities. Notebook outputs store values from the execution of their respective cells. To reproduce a calculation, specify the environment, restart the kernel, and execute the required calculation cells in order.

F.4 Recording and citing results

Attach the geometry, action variant, parameter values, and code version to results. Verify that CSV values, potential curves, and phase labels refer to the same inputs. Distinguish analytic results from numerical results on an input grid. The repository's LICENSE applies. Providing the procedure, inputs, and outputs together enables reproduction and comparison under the same conditions.

G Curvature in a noncoordinate frame

G.1 The structure-coefficient term

The exterior derivative of $\omega^a{}_b = \Gamma^a{}_{bc}e^c$ contains both coefficient derivatives and a term from de^c:

$$d\omega^a{}_b = d\Gamma^a{}_{bc} \wedge e^c + \frac{1}{2}\Gamma^a{}_{be}C^e{}_{cd}e^c \wedge e^d \tag{G-01}$$

Adding $\omega^a{}_e \wedge \omega^e{}_b$ and reading off two-form components gives the term $+\Gamma^a{}_{be}C^e{}_{cd}$ in (M2-04). Constant frame components do not remove this term in a noncoordinate frame. It vanishes for the torus, where $C = 0$, but contributes differently on S^3 and Nil, where $C \neq 0$.

G.2 Independence of signs, coefficients, and integration

The structure-coefficient term in curvature, the torsion-input dictionary, NY two-form coefficients, and global volume are defined separately. Metric compatibility alone does not determine the NY contraction coefficient, and agreement of local scalar curvatures does not determine the integration volume.

Likewise, $T = -\tau$ preserves the connection; it does not make FULL and REE the same action. The actions that coincide on the ansatz are the TT and positive-REE diagnostics. The FULL NY density vanishes as their density difference.

Acknowledgments

The author acknowledges the use of the following AI tools during manuscript preparation: Claude Opus 4.5 (Anthropic; accessed 2025-12), Gemini 3 Pro (Google; accessed 2025-12), Grok 4.1 Thinking (xAI; accessed 2025-12), and GPT-5.2 Thinking via ChatGPT (OpenAI; accessed 2025-12). These tools were used for language editing (including Japanese-English translation), outlining and improving exposition, drafting and reviewing auxiliary code (scripts and pseudocode) that was subsequently tested and validated by the author, and proposing candidate consistency checks and alternative derivations to be independently verified by the author. They were additionally used for literature discovery support (keyword generation and preliminary summaries of candidate papers); all references and factual claims were verified by the author using primary sources.

For this revision, Codex (OpenAI; accessed 2026-10) also assisted with computational and plotting code, bilingual presentation, and validation. The numerical values and plots were produced by deterministic evaluations of the stated formulas and checked against the accompanying implementation; they are not synthetic images produced by a generative image model. The author takes full responsibility for the content and for any remaining errors.

This work was developed within the informal collaborative project **DPPU** (**D**onut-like topology, **P**lanck-scale compactness, **P**recession dynamics, **U**niverse).

References

[1] F. W. Hehl, P. von der Heyde, G. D. Kerlick, and J. M. Nester, "General relativity with spin and torsion: Foundations and prospects," Rev. Mod. Phys. **48**, 393 (1976).

[2] O. Chandia and J. Zanelli, "Topological invariants, instantons, and the chiral anomaly on spaces with torsion," Phys. Rev. D **55**, 7580 (1997).

[3] H. T. Nieh and M. L. Yan, "An identity in Riemann-Cartan geometry," J. Math. Phys. **23**, 373 (1982).

[4] A. Corichi, I. Rubalcava-García, and T. Vukašinac, "Actions, topological terms and boundaries in first order gravity: A review," Int. J. Mod. Phys. D **25**, 1630011 (2016), arXiv:1604.07764.

[5] C. LeBrun, "Einstein metrics, complex surfaces, and symplectic 4-manifolds," Math. Proc. Cambridge Philos. Soc. **147**, 1–8 (2009), doi:10.1017/S0305004109002345.